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DSA :-

1) find out all the list of companies which are conducting placement drive using DSA.

2) what is the syllabus of DSA in GATE exam.

Variables in

Amazon site :-

- 1) colour (string)
- 2) price (double)
- 3) currency-code (string)
- 4) Discount-percentage (float)
- 5) size (string)
- 6) Address (string)
- 7) Name (string)
- 8) product-ID (string)
- 9) Brand (string)
- 10) Quantity (int)
- 11) Performance-Rating (float)
- 12) Delivery-Date (DateTime)
- 13) Is-In-Stock (boolean)
- 14) weight (float)

- 15) order-status (string)
- 16) Manufacturing-Address (string)
- 17) phone-Number (unsigned long int)
- 18) User-ID (string)
- 19) Cloth-Material (string)
- 20) Product-Details (class)
- 21) Cart (class)
- 22) email (string)
- 23) Items (list)
- 24) Order-ID (string)
- 25) overall-amount (double)

Ticket Variables:-

Ticket-id (string)
Date (DateTime)
Source } string
Destination }
Price (Double)
PNR-No. (string)
Seat-Number (string)
Booking-status (string)

Distance (float)
Bus-Service-Name (string)
Bus-No. }
Person-Name } string
Age (int)
Gender (char)
Departure-time } time
Arrival-time }
Distance (float)

3) find out what are the set of variables & data type in Amazon invoice mail.

4) Gas bill variables

5) Current bill variables

1) List of companies:-

→ Global Tech & Product Companies:-

Google

Amazon

Microsoft (IDC, IT)

Uber

Adobe

Oracle

Atlassian

Intuit

Samsung R & D.

JP Morgan

E-commerce,

→ Fin Tech & Indian startups:-

Flipkart

Mynta

PhonePe

Swiggy

Razorpay

CRED

Walmart

Global Technology

→ IT Services & Consulting :-
TCS, Tata Consultancy Services)
Infosys
Wipro
Accenture
Cognizant
IBM
Deloitte

2) Syllabus of DSA in GATE exam :-
GATE CS paper :-

i) programming & Data structures :-

→ programming in C : Basic concepts, functions, recursion & parameter passing.
→ Data structures : Arrays, stacks, queues, linked lists, trees, binary search trees (BST), binary heaps, and graphs.

ii) Algorithms :-

→ Algorithm Analysis :- Asymptotic Notation (O , Ω , Θ) and worst / average case time & space complexity analysis.

→ searching and sorting : Linear search, Binary search, sorting Algorithms (Merge sort, quick sort, selection sort, bubble sort, and insertion sort).

→ Algorithm Design Paradigms : Greedy Algorithms, divide and conquer, and dynamic programming.

→ Graph Algorithms : Traversals (Breadth-First search and Depth-First search), Minimum spanning Trees (Kruskal's & Prim's) & shortest path Algorithms (Dijkstra's & Bellman-Ford).

→ Hashing & Basic Complexity: Hashing Techniques & an Introduction to NP-Completeness.

3) Variables in Amazon Invoice mail:-

invoice-id (string)	items (List)
order-id (string)	product-name (str)
invoice-date (int)	Quantity (int)
customer-name (string)	Total (Double)
Address (string)	tax (Double)
email (string)	shipping-fee (Double)
	each-item-price (double)

4) current bill:-

bill-id (string)	Address (string)
customer-id (string)	District (string)
meter-num (string)	user-name (string)
reading-date (int)	AVG_UNITS (int)
volume-used (Double)	BILLED_UNITS (int)
net-amount (Double)	Loss / Gain (double)
govt subsidy-amount (Double)	phone number
payment-status (Boolean)	assigned long int)

5) Gas bill :-

Consumer name (string)
Consumer No. (unsigned long int).

LPD ID (unsigned long long int).

Address (string).

Mobile No. (unsigned long int).

Billing - Date (date)
Due - 4 Date (date)

6	9
1	32
2	40

Invoice -

Number (string)

Total Amount (double)